MA1521 Calculus for Computing

Lecture Group 2

Lecturer: Prof To Wing Keung

Lesson 14: Friday 02 October 2026

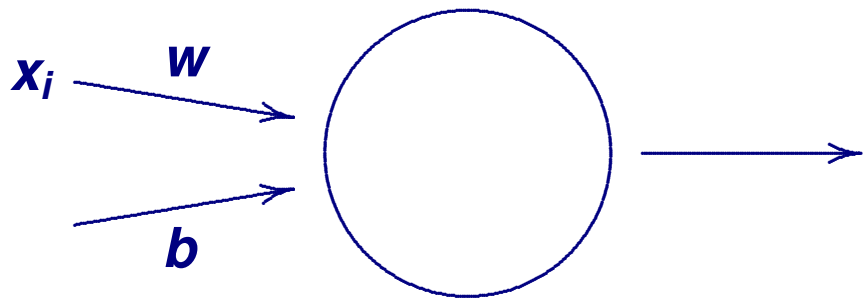
(Week 7)



A glimpse into training of neural networks in Machine Learning:

A demonstration app on training one neuron (constructed by my colleague Dr Wong Tin Lok):

https://wongtinlok.github.io/descent/



Traininng data: n data points (x_i, y_i), $i = 1, \cdots, n$.

- x_i is the net asset of the i-th person,
- y_i is the declaration of the i-th person on whether he/she is rich (**1**) or poor (**0**).

For the neuron, its activation function is given by

$$\varphi(s) = \frac{1}{1 + e^{-s}}, \qquad 0 < \varphi(s) < 1$$

so that for each input x_i, the output of the neuron is:

$$\varphi(x_i w + b) = \frac{1}{1 + e^{-(x_i w + b)}}.$$

Here the parameter w is the weight attached to the input, and the parameter b is called the bias.

The objective is to train the neuron so that its output will become closer and closer to human declaration.

Discrepancy function/Cost function:

$$C(w,b) = \sum_{i=1}^{n} (\varphi(x_i w + b) - y_i)^2$$

$$= \sum_{i=1}^{n} \left(\frac{1}{1 + e^{-(x_i w + b)}} - y_i\right)^2$$

The parameters (w, b) will be modified after each training, so that $C(w, b)$ will become smaller.

Direction of steepest descent: From MA1521, we know that after each training, (w, b) should move in the direction of $-\frac{\nabla C(w,b)}{\|C(w,b)\|}$ so that the cost function will decrease most rapidly among the various directions.

Double integrals.

In Calculus of one variable, our attempt to find the area under the curve $y = f(x)$ led to the definition of the definite integral.

We now seek to find volume under a surface $z = f(x, y)$ and in the process we arrive at the definition of a double integral.

We start by reviewing how we arrive at the definite integral of functions of a single variable:

Step 1. Suppose $f(x)$ is defined for $a \leq x \leq b$. We divide the interval $[a, b]$ into n subintervals of equal size $\triangle x = \dfrac{b-a}{n}$.

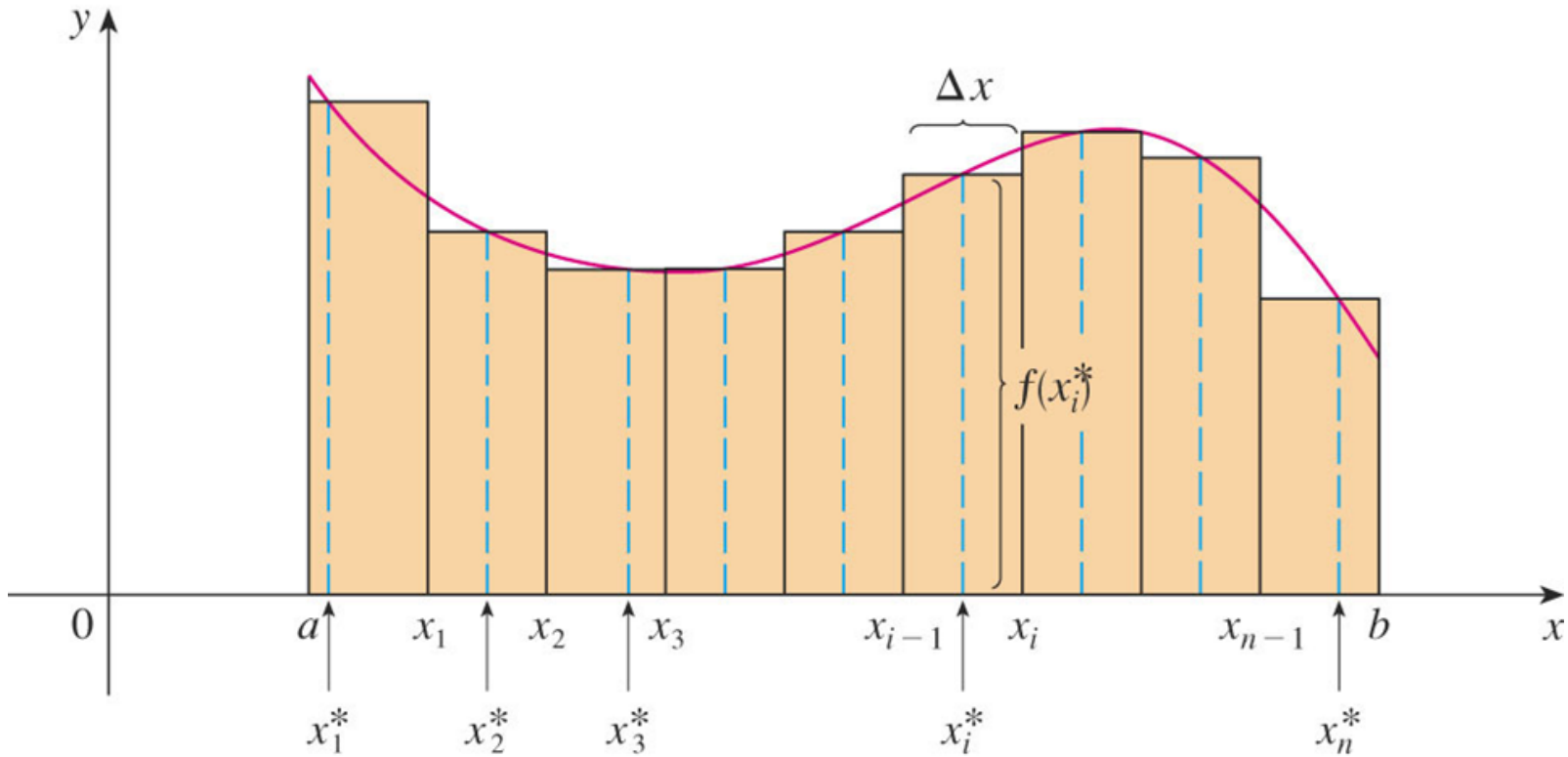
Step 2. We choose sample points x_i^* from these subintervals and form the **Riemann Sum**

$$\sum_{i=1}^{n} f(x_i^*) \triangle x.$$

Step 3. Take the limit of such sum as $n \to \infty$ to obtain the definite integral of f from a to b:

$$\int_a^b f(x)dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*) \triangle x.$$

In the special case where $f(x) \geq 0$, the integral $\int_a^b f(x)dx$ represents the area under the curve ~~$f(x)$~~ $y = f(x)$ from a to b.



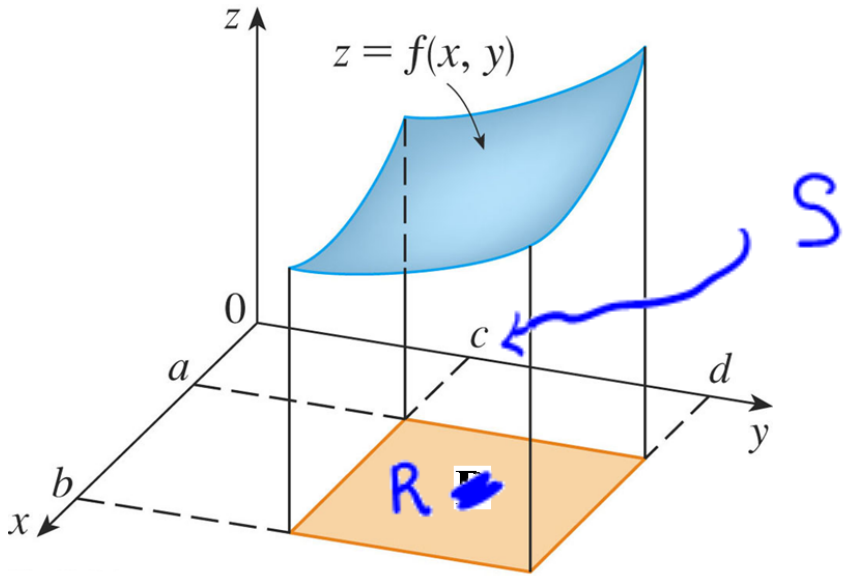
Double Integrals

Suppose $f(x, y)$ is defined on a closed rectangle

$$R = [a, b] \times [c, d] = \{(x, y) \in \mathbb{R}^2 : a \leq x \leq b, c \leq y \leq d\}.$$

Suppose $f(x, y) \geq 0$. The graph of f is the surface $z = f(x, y)$ above the region R.

Let S be the solid that lies above R and under the graph of f. How can we find the volume of S?



We mimic the process of defining definite integrals:

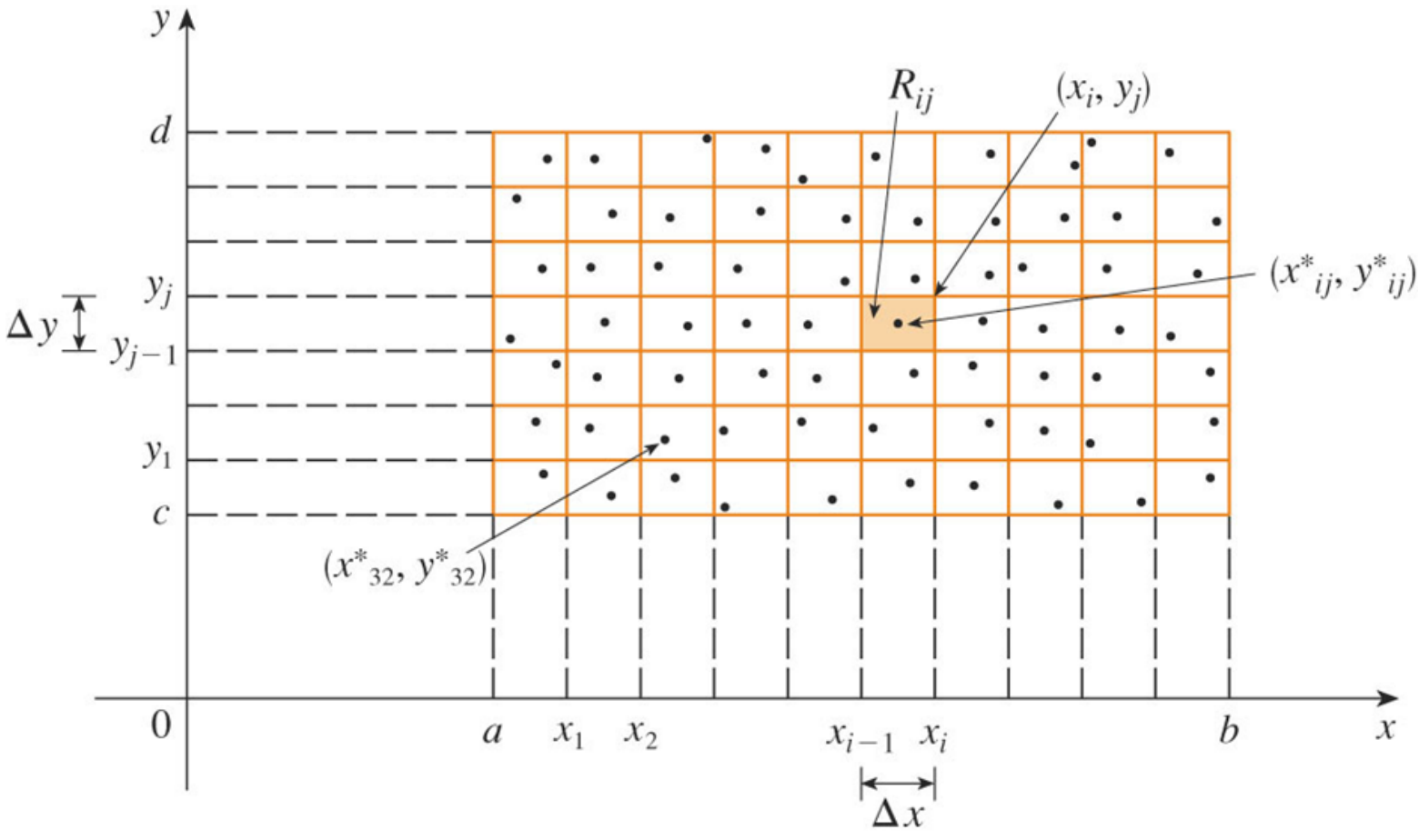
Step 1. Divide R into subrectangles. We do this by

- dividing the interval $[a, b]$ into m subintervals $[x_{i-1}, x_i]$ of equal length $\triangle x = \dfrac{b-a}{m}$, and
- dividing the interval $[c, d]$ into n subintervals $[y_{j-1}, y_j]$ of equal length $\triangle y = \dfrac{d-c}{n}$.

Form subrectangles

$$R_{ij} = [x_{i-1}, x_i] \times [y_{j-1}, y_j], \quad \text{for all } 1 \leq i \leq m, 1 \leq j \leq n.$$

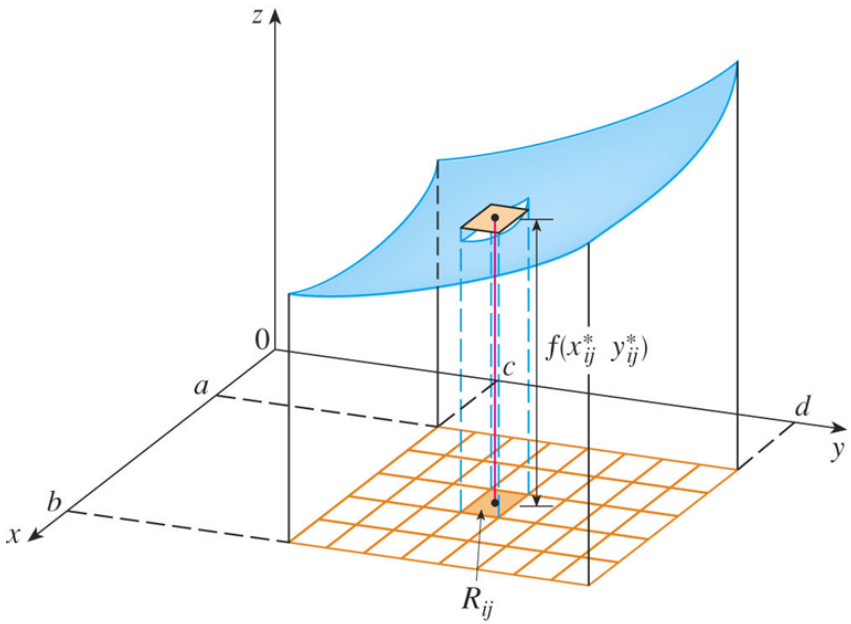
Each of these subrectangles has area $\triangle A = \triangle x \triangle y$.

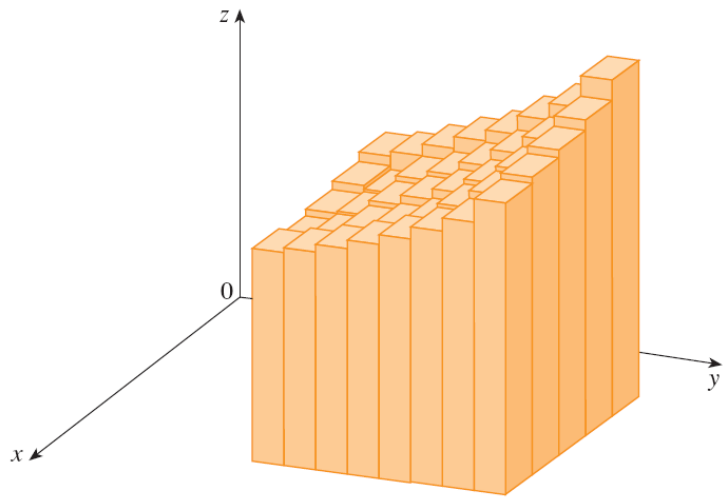

y
d
y_j
Δy
y_{j-1}
y_1
c
0
a
x_1
x_2
x_{i-1}
x_i
b
x
Δx
R_{ij}
(x_i, y_j)
(x^*_{ij}, y^*_{ij})
(x^*_{32}, y^*_{32})

Step 2. Choose a point (x_{ij}^*, y_{ij}^*) in each R_{ij}.

Then approximate the part of S lies above R_{ij} by a thin rectangle box with base R_{ij} and height $f(x_{ij}^*, y_{ij}^*)$.

The volume of this box is given by $f(x_{ij}^*, y_{ij}^*)\triangle A$.



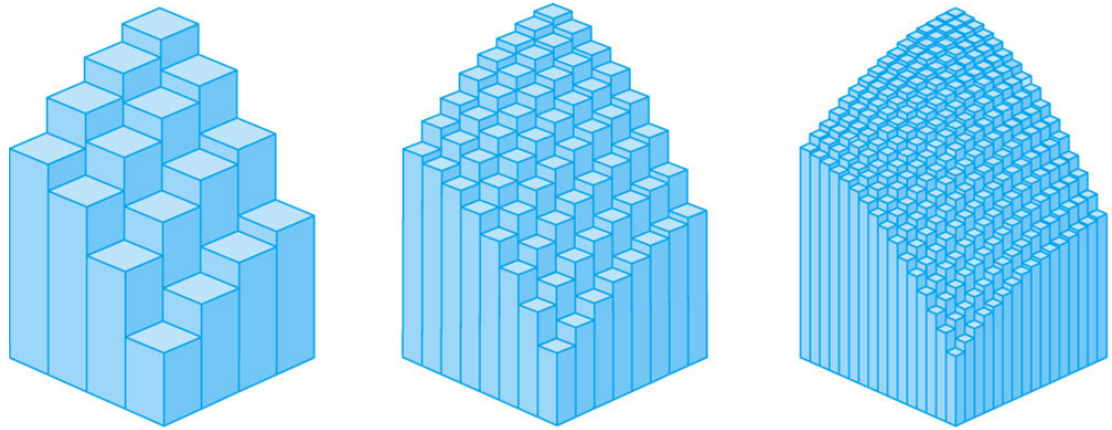


Upon adding (i.e. taking the Riemann sum of) the volumes of all these thin boxes, we get an approximation of the total volume V of S:

$$V \approx \sum_{i=1}^{m} \sum_{j=1}^{n} f(x_{ij}^*, y_{ij}^*) \triangle A.$$

Step 3: For nice functions $f(x, y)$, the approximation becomes better as $m, n \to \infty$. So we get

$$V = \lim_{m,n \to \infty} \sum_{i=1}^{m} \sum_{j=1}^{n} f(x_{ij}^*, y_{ij}^*) \triangle A.$$



(a) $m = n = 4$, $V \approx 41.5$ (b) $m = n = 8$, $V \approx 44.875$ (c) $m = n = 16$, $V \approx 46.46875$

Definition. The **double integral** of f over the region R is

$$\iint_R f(x,y)\,dA = \lim_{m,n\to\infty} \sum_{i=1}^{m}\sum_{j=1}^{n} f(x_{ij}^*, y_{ij}^*)\triangle A.$$

Informally, we think of dA as the area of a small subregion in R, so that $f(x,y)\,dA$ is the volume of the 'slender/tubular' solid under the graph of $f(x,y)$ and over the small subregion. Then the double integral sign $\iint_R$ means we take the limit of Riemann sum of the volumes of the 'slender/tubular' solids.

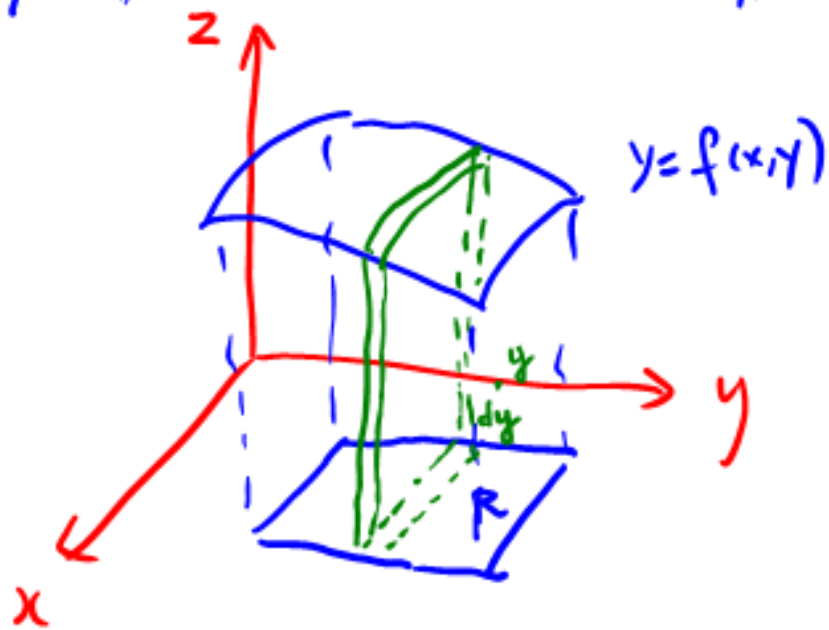
Summary: If $f(x,y) \geq 0$, the volume V of the solid region that lies above the region R and below the surface $z = f(x,y)$ is

$$V = \iint_R f(x,y)\,dA.$$

Fubini's Theorem (First Form).

Suppose $f(x, y)$ is continuous over the rectangular region $R : a \leq x \leq b,\ c \leq y \leq d$. Then

$$\iint_R f(x, y)\, dA \overset{(1)}{=} \int_{y=c}^{y=d} \left(\int_{x=a}^{x=b} f(x, y)\, dx \right) dy \overset{(2)}{=} \int_{x=a}^{x=b} \left(\int_{y=c}^{y=d} f(x, y)\, dy \right) dx.$$



$$= \int_{x=a}^{x=b} CSArea(y)\, dx$$

Informal Justification of (1): For a given y,

$$CSArea(y) = \int_a^b f(x, y)\, dx$$

is the area of a cross section of the solid S under the graph of $z = f(x, y)$ for that y.

Then $CSArea(y)\, dy$ is an approximate volume of a 'thin' cross sectional solid with thickness dy.

Thus $\int_c^d CSArea(y)\, dy = \int_c^d \int_a^b f(x, y)\, dx\, dy$ is the volume of the solid under the graph $z = f(x, y)$ over R.

The integrals

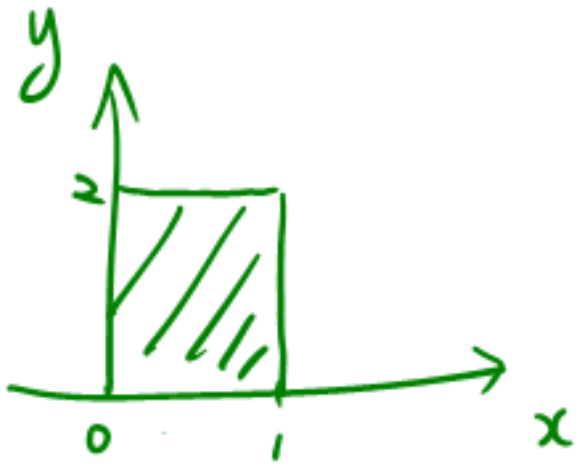
$$\int_c^d \left(\int_a^b f(x, y)\, dx \right) dy, \qquad \int_a^b \left(\int_c^d f(x, y)\, dy \right) dx$$

are called **iterated integrals.**

For both type of iterated integrals, we first compute the inner integrals (w.r.t. inner variable), getting an expression in the outer variable.

Then we proceed to evaluate the outer integral, getting finally a number as the answer.

Example. Evaluate the double integral $\iint_R (2x + 2y)\, dA$, where R is the region in the xy-plane bounded by the lines $x = 0$, $x = 1$, $y = 0$ and $y = 2$.



$$\iint_R (2x+2y)\, dA$$

$$= \int_{y=0}^{y=2} \left(\int_{x=0}^{x=1} (2x+2y)\, dx \right) dy$$

$$= \int_{y=0}^{y=2} \left[x^2 + 2yx \right]_{x=0}^{x=1} dy$$

$$= \int_{y=0}^{y=2} \left((1+2y) - (0+0) \right) dy$$

$$= \int_{y=0}^{y=2} (1+2y)\, dy = \left[y + y^2 \right]_{y=0}^{y=2} = 2+4-0 = 6.$$

$$\int_{x=0}^{x=1} \left(\int_{y=0}^{y=2} (2x+2y)\, dy \right) dx$$

$$= \int_{x=0}^{x=1} \left[2xy + y^2 \right]_{y=0}^{y=2} dx$$

$$= \int_{x=0}^{x=1} \left(4x+4-(0+0) \right) dx$$

$$= \int_{x=0}^{x=1} (4x+4)\, dx$$

$$= \left[2x^2+4x \right]_{x=0}^{x=1} = 2+4-(0+0) = \underline{\underline{6}}.$$

Double integral of $f(x, y)$ over general region D

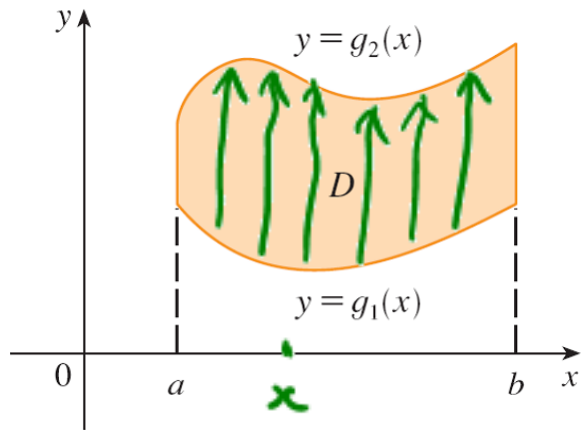
Fubini's Theorem (for Type I region)

If f is continuous on a Type I region D such that

$$D = \{(x, y) : a \leq x \leq b, \;\; g_1(x) \leq y \leq g_2(x)\}$$

then

$$\iint_D f(x, y)\, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy\, dx.$$



Tricky variable: y

Simple variable: x

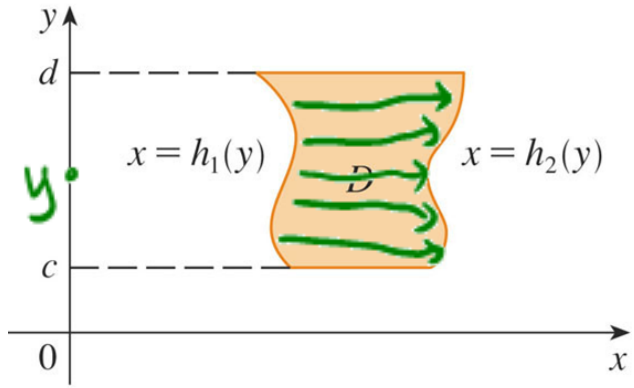
Fubini's Theorem (for Type II region)

If f is continuous on a Type II region D such that

$$D = \{(x, y) : c \leq y \leq d, \ \ h_1(y) \leq x \leq h_2(y)\}$$

then

$$\iint_D f(x, y)\, dA = \int_c^d \left(\int_{h_1(y)}^{h_2(y)} f(x, y)\, dx \right) dy.$$



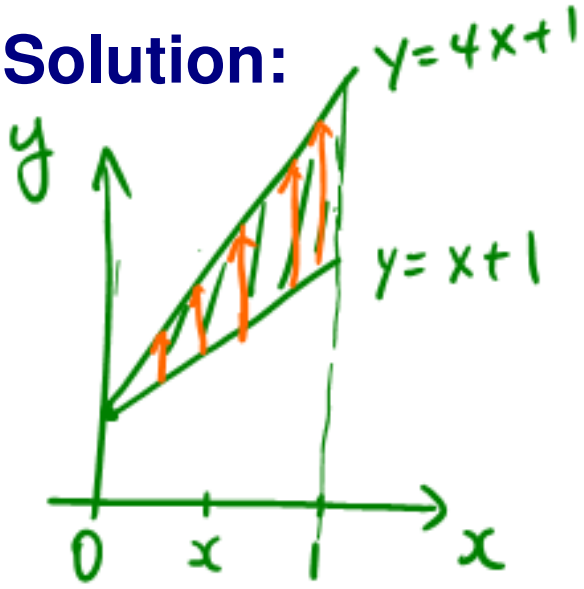
Tricky variable: x

simple variable: y

Example. Evaluate the double integral $\iint_R 2x\,dA$, where R is the region in the xy-plane given by

$$R = \{(x, y) \in \mathbb{R}^2 : 0 \le x \le 1,\ \ x + 1 \le y \le 4x + 1\}.$$

Solution:



Tricky variable: y

simple variable: x

$$\iint_R 2x\,dA$$

$$= \int_{x=0}^{x=1} \left(\int_{y=x+1}^{y=4x+1} 2x\,dy \right) dx$$

$$= \int_{x=0}^{x=1} \Big[2xy \Big]_{y=x+1}^{y=4x+1} dx$$

$$= \int_{x=0}^{x=1} \Big(2x(4x+1) - 2x(x+1) \Big) dx$$

$$= \int_{x=0}^{x=1} 6x^2\,dx = \Big[2x^3 \Big]_{x=0}^{x=1} = 2 - 0 = 2.$$

